

Remarks on the talk by Bodo Pareigis on 15 March 2000

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ABSTRACT. These remarks refer to a talk by Bodo Pareigis based on his joint work [HP] with Daniela Hobst.

We work in a 2-category $\mathcal{K}$ which admits the Eilenberg-Moore construction for monads and comonads.

Suppose $(\mathcal{A}, S)$ is a monad and $(\mathcal{A}, G)$ is a comonad in $\mathcal{K}$. A (mixed) distributive law of G over S is a 2-cell $\psi : S G \Rightarrow G S$ satisfying familiar compatibility conditions. This allows us to lift the comonad G on $\mathcal{A}$ to a comonad $\bar{G}$ on the Eilenberg-Moore-algebra object $\mathcal{A}^S$ for S . Then we also have the Eilenberg-Moore-coalgebra object $(\mathcal{A}^S)^{\bar{G}}$ for $\bar{G}$.

For simplicity, suppose $S : \mathcal{A} \rightarrow \mathcal{A}$ has a right adjoint $H : \mathcal{A} \rightarrow \mathcal{A}$. Then H inherits a comonad structure from the monad structure on S , the mixed distributive law $\psi : S G \Rightarrow G S$ corresponds to a distributive law $\lambda : G H \Rightarrow H G$ between the comonads G and H , and $G H$ becomes a comonad on $\mathcal{A}$ in the obvious way. Moreover, we have the identification

$$\mathcal{A}^H = \mathcal{A}^S$$

between the Eilenberg-Moore-coalgebra object for H and the Eilenberg-Moore-algebra object for S , and the identification

$$(\mathcal{A}^S)^{\bar{G}} = \mathcal{A}^{G H}.$$

It is a trivial remark that, for any comonad $(\mathcal{A}, G)$ with Eilenberg-Moore object

$$\begin{array}{c} \xrightarrow{V} \\ \mathcal{A}^G \perp \mathcal{A}, \\ \xleftarrow{E} \end{array}$$

we have the bijection

$$\mathcal{K}(\mathcal{A}^G, \mathcal{A})(V, V) \cong \mathcal{K}(\mathcal{A}, \mathcal{A})(G, 1_{\mathcal{A}})$$

since $G = V E$. Consequently, if $W : (\mathcal{A}^S)^{\bar{G}} \rightarrow \mathcal{A}$ is the canonical "underlying" arrow, we have bijections

$$(0.1) \quad \mathcal{K}((\mathcal{A}^S)^{\bar{G}}, \mathcal{A})(W, W) \cong \mathcal{K}(\mathcal{A}, \mathcal{A})(GH, 1_{\mathcal{A}}) \cong \mathcal{K}(\mathcal{A}, \mathcal{A})(G, S)$$

where the second bijection uses the adjunction $S \dashv H$.

Example (Hobst-Pareigis [HP]) Take $\mathcal{K}$ to be the 2-category $\mathcal{M}\text{-Cat}$ of categories enriched in the monoidal category $\mathcal{M}$. For simplicity we suppose that $\mathcal{M}$ is left closed, with internal hom denoted by $[X, Y]$, and has equalizers and coequalizers. Take A to be a monoid in $\mathcal{M}$, take C to be a comonoid in $\mathcal{M}$, and take $\psi : C \otimes A \rightarrow A \otimes C$ to be an "entwiner". Put $S = - \otimes A$ so that $H = [A, -]$, and put $G = - \otimes C$. The mixed distributive law $\psi : SG \Rightarrow GS$ can be identified with the entwiner. The Hobst-Pareigis category $\mathcal{M}_A^C(\psi)$ is precisely $(\mathcal{M}^S)^{\bar{G}}$. Observation 0.1 immediately gives the bijection

$$\mathcal{M}\text{-Cat}(W, W) \cong \mathcal{M}\text{-Cat}(G, S) \cong \mathcal{M}(C, A).$$

With a little more structure in our general theory we could actually obtain a monoid isomorphism

$$[\mathcal{M}_A^C(\psi), \mathcal{M}](W, W) \cong [C, A]$$

in $\mathcal{M}$.

Suppose $\mathcal{K}$ is a monoidal bicategory and that $\mathcal{A}$ is a monoidal object (that is, a pseudo-monoid in $\mathcal{K}$). Recent work [Mjk], [McC] shows that monoidal structures for the comonad G on $\mathcal{A}$ are in bijection with monoidal structures for $\mathcal{A}^G$ over $\mathcal{A}$. Consequently, monoidal structures on $(\mathcal{A}^S)^{\bar{G}}$ over $\mathcal{A}$ are in bijection with monoidal structures on GH . If we suppose S is an opmonoidal monad and G is a monoidal comonad then there is an obvious candidate for a monoidal structure on GH ; however, there are two conditions relating the opmonoidal structure on S , the monoidal structure on G , and the mixed distributive law ψ , in order that this really is a monoidal structure on GH .

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